Notes: Kaplan and Zingales (QJE, 1997)

Do Investment-Cash Flow Sensitivities Provide Useful Measures of Financing Constraints?

Contents

1 Big picture

- **Question.** Are investment–cash flow sensitivities a useful measure of financing constraints?
- **One-word answer. No.** The whole point of the paper is that a higher sensitivity of investment to cash flow cannot be read mechanically as evidence that a firm is more financially constrained.
- **Setting.** The paper goes back to the canonical FHP [1988] sample: **49** low-dividend manufacturing firms from **1970–1984**, the exact firms that generated the classic result that "financially constrained" firms have high investment–cash flow sensitivities.
- **Core challenge.** The existing literature does *not* directly verify the key assumption it needs. It assumes that the sensitivity of investment to internal funds rises monotonically with the degree of financing constraints. Kaplan and Zingales argue that this monotonicity is neither theoretically obvious nor empirically verified.
- **Main theoretical result.** In a simple model with costly external finance, constrained firms should in general react to internal funds, but *cross-firm differences* in that reaction need not be monotone in financing frictions. The sign depends on the curvature of the production function and the curvature of the external-finance cost function.
- **Main empirical strategy.** The authors do unusually direct measurement. They read annual reports and 10-Ks, management discussions of liquidity, financial-statement footnotes, and Wall Street Journal coverage to classify each firm-year into one of five categories: definitely not constrained, likely not constrained, possibly constrained, likely constrained, and definitely constrained.
- **Main descriptive result.** The famous FHP low-dividend firms are *usually not constrained*. Of all firm-years, **54.5%** are classified as NFC and **30.9%** as LNFC. So **85.3%** of firm-years show no evidence that financing constraints limited investment. Only **14.7%** of firm-years are even possibly constrained.
- **Firm-level punchline. 19 of the 49 firms** are never classified as constrained in any year. That is almost **40%** of the sample.
- **Main regression result.** The firms that look *less* constrained have *higher*, not lower, investment–cash flow sensitivities. Over the full sample, the coefficient for never-constrained firms is about **0.702**, versus **0.340** for likely constrained firms and **0.180** for possibly constrained firms. The full-sample coefficient is about **0.395**.
- **Robustness result.** The same pattern survives:
  - subperiod splits;
  - objective financial-health rules such as interest-coverage cutoffs and dividend restrictions;
  - regressions that use only *predetermined* lagged financial status;
  - cash-stock specifications;
  - outlier trimming;
  - Euler-equation specifications that avoid reliance on Tobin's Q.
- **Why this matters.** A very large literature interprets high investment–cash flow sensitivity as evidence of tight financing constraints and then uses that statistic to infer which institutions or policies alleviate those constraints. KZ argue that this interpretation is not secure.

- **Economic takeaway.** The paper does *not* say financing constraints do not exist. It says the usual statistic is not a monotone sufficient statistic for them. Some very healthy firms appear to invest mainly out of internal funds despite having ample access to cheap outside finance.

2 Data and measurement (key objects)

2.1 The FHP sample and why KZ use it

- The sample is the **49 Class 1 firms** from FHP [1988].
- FHP define these as firms with a dividend payout ratio below **10%** in at least **10 of the 15** sample years.
- For comparison:
  - **39** firms are Class 2 (dividend payout between 10% and 20%);
  - **334** firms are Class 3 (the remaining firms).
- KZ choose the Class 1 firms for three reasons:
  - these firms are exactly the firms that exhibit strong investment–cash flow sensitivities;
  - FHP explicitly argue they are financially constrained;
  - the sample is small enough for firm-by-firm archival work.

2.2 Primary data sources

- Annual reports and 10-K filings for each firm-year.
- Letters to shareholders.
- Management discussion of operations and liquidity.
- Financial statements and footnotes.
- *Wall Street Journal Index* entries across the fifteen-year sample.
- Standard accounting variables from COMPUSTAT, except for Coleco, which the authors code directly from annual reports.

Why this matters. The paper is not just re-running regressions on COMPUSTAT. It tries to observe directly whether firms could have obtained funds and whether managers themselves said liquidity was limiting investment.

2.3 Sample period and usable observations

- Sample period: **1970–1984**.
- Usable firm-year observations in KZ: **719**.
- This is slightly below the FHP total because of fiscal-year changes, missing SEC filings, and missing stock prices.

2.4 Main empirical objects

The basic investment equation uses:

$$\frac{\mathrm{I}_{it}}{\mathrm{K}_{i,t-1}} = \text{capital expenditures/beginning-of-year capital}, \tag{1}$$
$$\frac{\mathrm{CF}_{it}}{\mathrm{K}_{i,t-1}} = \frac{\text{earnings before extraordinary items} + \text{depreciation}}{\text{beginning-of-year capital}}, \tag{2}$$
$$Q_{i,t-1} = \frac{\mathrm{MVAssets}_{i,t-1}}{\mathrm{BVAssets}_{i,t-1}}. \tag{3}$$

with

$$\mathrm{K}_{i,t-1} = \text{net property, plant, and equipment at the beginning of the fiscal year.}$$

Interpretation.

- Investment is COMPUSTAT item 128.
- Cash flow is item 18 plus item 14.
- Capital is item 8 at the start of the year.
- Q is a market-to-book asset measure computed at the beginning of the fiscal year.

The paper defines market value of assets as

$$\text{MVAssets} = \text{BVAssets} + \text{MVCommon Equity} - \text{BVCommon Equity} - \text{Deferred Taxes}. \tag{4}$$

Interpretation. KZ use a simpler but arguably cleaner market-to-book Q than FHP, who use a more elaborate replacement-cost construction.

2.5 Quantitative measures of financial health

The paper uses several financial-health objects to validate the qualitative classification:

$$\text{Interest Coverage}_{it} = \frac{\mathrm{EBITDA}_{it}}{\text{Interest Expense}_{it}}, \tag{5}$$
$$\text{Debt Ratio}_{it} = \frac{\mathrm{Debt}_{it}}{\text{Total Capital}_{it}}, \tag{6}$$
$$\text{Cash Ratio}_{it} = \frac{\mathrm{Cash}_{it}}{\mathrm{K}_{i,t-1}}, \tag{7}$$
$$\text{Unused Line Ratio}_{it} = \frac{\text{Unused Credit Line}_{it}}{\mathrm{K}_{i,t-1}}, \tag{8}$$
$$\text{Slack Ratio}_{it} = \frac{\mathrm{Cash}_{it} + \text{Unused Credit Line}_{it}}{\mathrm{K}_{i,t-1}}. \tag{9}$$

Interpretation.

- Interest coverage says whether operations comfortably service interest.
- Debt ratio says how levered the firm is.
- Cash, unused lines, and slack say how much immediately available liquidity the firm has.

The authors also look at:

- whether debt covenants restrict dividend payments;
- unrestricted retained earnings available for dividends;
- acquisitions as a share of capital;
- sales growth and inventory growth;
- changes in debt and equity issuance.

2.6 What the five financing-status categories mean

- **NFC (not financially constrained).** The firm increases or initiates dividends, repurchases stock, explicitly states it has excess liquidity, or otherwise clearly has more funds than it needs for investment.
- **LNFC (likely not financially constrained).** The firm looks healthy and liquid, with strong coverage, cash reserves, or unused lines of credit, but without the especially strong evidence used for NFC.
- **PFC (possibly financially constrained).** The firm is ambiguous: not obviously strapped for cash, but not obviously liquid either.
- **LFC (likely financially constrained).** The firm reports financing problems, postpones a security issue because of bad market conditions, cuts dividends, or otherwise looks short of liquidity.
- **FC (financially constrained).** The firm violates debt covenants, loses its normal credit access, renegotiates debt, or explicitly says it must reduce investment because of financing problems.

Why the authors think this is informative.

- After 1977, SEC Regulation S-K requires explicit discussion of liquidity and capital resources.
- The authors do not rely only on managerial narrative; they combine it with financial statements and footnotes.
- Any tendency for managers to hide bad news should bias the classification *against* finding constrained firms.

2.7 How many firm-years fall into each bucket?

Using the **719** firm-years:

- NFC: **393** firm-years = **54.5%**.
- LNFC: **221** firm-years = **30.9%**.

- PFC: **52** firm-years = **7.3%**.
- LFC: **34** firm-years = **4.8%**.
- FC: **19** firm-years = **2.6%**.

So NFC+LNFC account for **85.3%** of firm-years.

2.8 What the validation data say

Table III shows strong monotonic patterns across annual categories.

- Median cash flow over capital falls from **0.506** in NFC years to **0.020** in FC years.
- Median (CF − I)/K falls from **0.110** in NFC years to about **-0.198** in FC years.
- Median interest coverage falls from **7.97** in NFC years to **1.09** in FC years.
- Median debt-to-total-capital rises from about **0.296** to **0.565**.
- The fraction of years with dividend restrictions rises from **6.1%** to **78.9%**.
- Median slack falls from about **0.725** of capital in NFC years to about **0.229** in FC years.

Interpretation. The qualitative ranking lines up with ordinary financial-health measures. It is not random.

2.9 Overall status over the whole sample

To compare investment–cash flow sensitivities over long windows, the paper aggregates firm-years into overall firm status:

- **Never constrained: 19 firms**. These are NFC or LNFC in every year.
- **Possibly constrained: 8 firms**. These are PFC in at least one year and never LFC/FC.
- **Likely constrained: 22 firms**. These are LFC or FC in at least one year.

The paper also constructs two subperiod classifications:

$$\text{1970–1977} \quad \text{and} \quad \text{1978–1984}.$$

Why subperiods?

- financing status can change over time;
- firms were typically smaller earlier in the sample;
- disclosures are more informative after 1977;
- FHP themselves show stronger sensitivities early in the sample.

2.10 An important fact from the overall-status table

Table XII shows that firms labeled "possibly constrained" are *not* obviously unhealthy over the full sample:

- median interest coverage is about **9.93** for possibly constrained firms versus **8.07** for never constrained firms and **4.84** for likely constrained firms;
- median cash-flow and Q values for possibly constrained firms are also not weak.

Interpretation. The middle category is genuinely ambiguous, not a pool of distressed firms hiding in the data.

3 Models and empirical specifications (all equations + interpretation)

3.1 What "financially constrained" means in the model

The paper defines a financing constraint as a wedge between the cost of internal funds and the cost of external funds.

- Let W denote internal funds available to the firm.
- Let E denote external funds raised.
- Let k denote the severity of the wedge between internal and external finance.

A larger k means a firm is *more* financially constrained.

3.2 The one-period investment problem

The firm chooses investment I to maximize profits:

$$\max_{I}\ F(I) - C(E,k) - I \quad \text{subject to} \quad I = W + E. \tag{10}$$

where

- $F'(I) > 0$ and $F''(I) < 0$;
- $C(E,k)$ is the deadweight cost of raising outside funds;
- C rises with the amount of outside finance E and with the severity of financing frictions k;
- C is assumed convex in E.

Interpretation.

- The production side says investment has diminishing returns.
- The finance side says external funds are costly because of asymmetric information, agency problems, or similar frictions.

- The whole debate is about whether the sensitivity of I to internal funds W can rank firms by k.

3.3 First-order condition and the basic comparative statics

Substituting $E = I - W$, the first-order condition is

$$F_I(I) = 1 + C_1(I - W, k). \tag{11}$$

Implicit differentiation gives

$$\frac{dI}{dW} = \frac{C_{11}}{C_{11} - F_{11}} > 0. \tag{12}$$

and

$$\frac{dI}{dk} = -\frac{C_{12}}{C_{11} - F_{11}} < 0 \quad \text{if } C_{12} > 0. \tag{13}$$

Interpretation.

- Equation (12): in an imperfect capital market, more internal funds raise investment.
- Equation (13): a more severe financing wedge lowers investment.
- These two propositions are *not* the controversial part.

3.4 Why monotonicity of investment–cash flow sensitivity is not guaranteed

The literature usually compares $\frac{dI}{dW}$ across groups of firms. That only makes sense if

$$\frac{dI}{dW}$$

moves monotonically with financial constraints.

But

$$\frac{d^2 I}{dW^2} = \frac{C_{11}^2 F_{111} - C_{111} F_{11}^2}{(C_{11} - F_{11})^3}. \tag{14}$$

Interpretation.

- The sign of $\frac{d^2 I}{dW^2}$ is *not* pinned down in general.
- It depends on third derivatives of both the production function and the external-finance cost function.
- So the sensitivity of investment to internal funds need *not* rise when financing becomes tighter.

The paper's key example is that if $C(\cdot)$ is quadratic and $F(I) = I^p$ with $0 < p < 1$, then $F_{111} > 0$ and the investment–cash flow sensitivity can actually *increase* with internal liquidity.

Main lesson. Even in a simple one-period model, a higher cash-flow coefficient is not a theoretically clean ranking of financing constraints.

3.5 Why the multi-period case is even harder

- In dynamic models, precautionary savings motives complicate the mapping even further.
- The paper cites Gross [1995], who obtains a nonmonotonic relationship between financing constraints and investment–cash flow sensitivities in a dynamic setting.
- The authors also note that overly conservative or risk-averse managers may prefer to finance investment out of internal cash even when external funds are available at low cost.

3.6 Baseline FHP-style investment regression

The empirical benchmark is the standard fixed-effects Q-investment regression:

$$\frac{\mathrm{I}_{it}}{\mathrm{K}_{i,t-1}} = \alpha_i + \eta_t + \beta Q_{i,t-1} + \gamma \frac{\mathrm{CF}_{it}}{\mathrm{K}_{i,t-1}} + u_{it}. \tag{15}$$

Interpretation.

- α_i are firm fixed effects.
- η_t are year effects.
- β controls for investment opportunities proxied by Q.
- γ is the famous investment–cash flow sensitivity.

3.7 How the paper tests the literature's monotonicity claim

The simplest tests run equation (15) separately on groups of firms:

- never constrained;
- possibly constrained;
- likely constrained;
- and analogous groupings in each subperiod.

Interpretation. If the literature's assumption were correct, γ should be highest for the most constrained group. The paper finds the opposite.

3.8 Predetermined annual-status interaction regression

To avoid using future information, the paper also uses lagged annual status:

$$\frac{\mathrm{I}_{it}}{\mathrm{K}_{i,t-1}} = \alpha_i + \eta_t + \beta Q_{i,t-1} + \gamma_0 \frac{\mathrm{CF}_{it}}{\mathrm{K}_{i,t-1}} + \sum_{g \in \{\mathrm{LNFC,PFC,LFC,FC}\}} \gamma_g D^g_{i,t-1} \frac{\mathrm{CF}_{it}}{\mathrm{K}_{i,t-1}} + u_{it}, \tag{16}$$

where the omitted base category is NFC.

Interpretation.

- γ_0 is the sensitivity for previously NFC firm-years.
- The interaction terms show whether other groups have higher or lower sensitivities than that benchmark.
- This is the paper's cleanest specification because financial status is measured using information available at the start of the year.

3.9 Validation regression for the financing-status indicators

Because theory unequivocally predicts that tighter financing constraints should lower investment levels, the paper validates its lagged indicators with

$$\frac{\mathrm{I}_{it}}{\mathrm{K}_{i,t-1}} = \alpha_i + \eta_t + \beta Q_{i,t-1} + \sum_{g \in \{\mathrm{LNFC},\mathrm{PFC},\mathrm{LFC},\mathrm{FC}\}} \delta_g D^g_{i,t-1} + u_{it}, \tag{17}$$

again using NFC as the omitted category.

Interpretation.

- If the classification is meaningful, the δ_g terms should decline as financing status worsens.
- That is exactly what the paper finds.

3.10 Ordered logit for classification validity

The paper also estimates an ordered logit of the form

$$\mathrm{Status}^*_{it} = X'_{it}\lambda + \varepsilon_{it}, \tag{18}$$

where observed status moves from NFC to FC as the latent index crosses cutoff points.

Interpretation. This is not the main economic equation. It is a validity check: do standard measures of financial health line up with the qualitative ranking? They do.

3.11 Cash-stock and internal-funds specifications

Because some of the literature focuses on beginning-of-period cash rather than contemporaneous cash flow, the paper also estimates specifications with

$$\frac{\mathrm{I}_{it}}{\mathrm{K}_{i,t-1}} = \alpha_i + \eta_t + \beta Q_{i,t-1} + \theta \frac{\mathrm{Cash}_{i,t-1}}{\mathrm{K}_{i,t-1}} + \phi \frac{\mathrm{CF}_{it}}{\mathrm{K}_{i,t-1}} + \text{status interactions} + u_{it}. \tag{19}$$

and with internal funds measured as

$$\frac{\mathrm{Cash}_{i,t-1} + \mathrm{CF}_{it}}{\mathrm{K}_{i,t-1}}. \tag{20}$$

Interpretation. The paper's main message survives when internal funds are measured more broadly.

3.12 Euler-equation robustness

The paper also implements a Bond–Meghir-style Euler-equation regression that includes lagged investment, lagged investment squared, sales, cash flow, and debt-squared terms.

Interpretation.

- This approach does not rely on Tobin's Q.
- So if the results survive, the "cash flow is just proxying for mismeasured Q" objection becomes much weaker.

4 Identification and instruments (what makes the estimates credible)

4.1 What the paper is actually identifying

- The paper is *not* testing whether capital-market frictions exist.
- It is testing whether investment–cash flow sensitivities rank firms monotonically by the severity of those frictions.
- That distinction is crucial. The literature often treats the ranking result as automatic. KZ say it is not.

4.2 Why direct observation is the paper's main identification strategy

- Instead of relying on theoretical priors like dividend payout, leverage, or bank affiliation, the paper reads what firms actually say about liquidity and financing.
- This is especially valuable after SEC Regulation S-K, which requires disclosure of liquidity trends, funding needs, and capital-resource plans.
- The archival evidence is combined with quantitative balance-sheet data.

Bottom line. The identification comes from a more direct observation of financing conditions than prior work used.

4.3 Why the sample choice is hard on the authors' own hypothesis

- The authors deliberately choose the classic FHP low-dividend firms.
- These are exactly the firms for which the conventional interpretation should work best.
- So if monotonicity fails even here, that is a serious problem for the broader literature.

4.4 Quantitative validation of the qualitative classifications

The paper does not stop at the narrative classification.

- In Table III, cash flow, net cash flow, interest coverage, cash, slack, and freedom to pay dividends all deteriorate as status moves from NFC to FC.
- In Table IV's ordered logits, the likelihood of being classified as constrained rises with debt and dividend restrictions and falls with cash flow, cash holdings, unrestricted retained earnings, and unused credit lines.

Why this matters. It is difficult to dismiss the categories as arbitrary when ordinary financial-health variables line up so cleanly with them.

4.5 Why lagged status matters

A natural objection is that firms with high investment–cash flow sensitivities may avoid becoming constrained simply because they only invest when cash is available. That would make within-sample classifications partly an outcome of financing policy.

The paper addresses this by using only *lagged* annual status in the key interaction regressions.

Interpretation. This makes the timing cleaner: last year's financing condition predicts this year's investment sensitivity.

4.6 Why Q mismeasurement is not a convincing alternative explanation

- A classic concern is that cash flow proxies for investment opportunities not captured by Q.
- The paper therefore estimates Euler-equation specifications that do not rely on Q.
- The result is unchanged: least constrained firms still have the highest cash-flow coefficients.

4.7 Why outliers are not driving the cross-sectional result

- The paper removes firm-years with sales growth above 30% and also removes firm-years in which net property, plant, and equipment more than doubles.
- The levels of the coefficients fall, but the ranking remains the same: unconstrained firms continue to show the highest sensitivities.

Important nuance. The outliers *do* matter for the original FHP result comparing low-dividend and high-dividend firms. Once high-growth outliers are trimmed, the low-dividend sample's overall sensitivity falls toward the high-dividend benchmark.

4.8 Why the "maybe these are just distressed firms" story does not work

The paper explicitly argues that its constrained firms are not simply distressed firms forced to divert cash to debt repayment.

- In constrained years, firms tend to *increase* debt rather than pay it down.
- Median interest coverage is about **4.20** for PFC years and **2.84** for LFC years.
- Over the full sample, likely constrained firms have median coverage of about **4.84**, which is weak relative to healthy firms but not a picture of pervasive insolvency.

4.9 Why previous sorting criteria are theoretically ambiguous

A major part of the paper's broader argument is that many priors used in the literature do not rank firms unambiguously.

- **Main-bank relationships:** could reduce information asymmetry and ease finance, or could increase hold-up and make finance more expensive.
- **Low leverage:** could mean spare debt capacity and easy access to finance, or could mean firms avoid leverage because distress costs are high.
- **High cash holdings:** could mean abundant liquidity, or precautionary savings by firms that expect future financing trouble.

Interpretation. If the priors are ambiguous, then a higher cash-flow coefficient cannot settle the question by itself.

4.10 Main limitations

- The paper studies one prominent but selected sample of **49** manufacturing firms.
- The classification has an unavoidable qualitative component, even though it is heavily validated.
- The paper undermines the monotonicity assumption, but it does not provide a universal replacement statistic for financing constraints.
- The broader claim that much of the literature should be reinterpreted is persuasive, but still partly extrapolative.

4.11 Relation to the literature

- Direct target: FHP [1988] and the investment–cash flow literature built on it.
- Related methodological alternatives: Bond and Meghir [1994], Whited [1992], Hubbard, Kashyap, and Whited [1995].
- Related substantive applications: Hoshi, Kashyap, and Scharfstein [1991]; Houston and James [1995]; Kashyap, Lamont, and Stein [1994]; Calomiris-related papers.

The paper's claim about its place in the literature. This is not just another application. It is the first paper to test directly the key monotonicity assumption on which that literature relies.

5 Tables and figures

General note. This is a very table-driven paper. There are no central figures doing the heavy lifting. The argument is built almost entirely from tables and from the contrast between direct archival classification and regression coefficients.

5.1 Table I: replication of the basic FHP result

Read. This table re-estimates the standard investment equation for the low-dividend FHP sample using KZ's data definitions.

- The table shows that KZ do recover the classic qualitative fact: these firms have a strong positive investment–cash flow sensitivity.
- Their cash-flow coefficients are somewhat smaller than FHP's, while their Q coefficients are larger.
- The authors attribute this mainly to a different and, in their view, cleaner Q measure.

Why it matters. This table is essential because it shows the authors are not overturning the literature by changing the sample. They start from the same famous sample and replicate the headline pattern.

TABLE I

COMPARISON OF REGRESSION OF INVESTMENT ON CASH FLOW AND Q WITH FAZZARI, HUBBARD, AND PETERSEN RESULTS

Regression of investment on cash flow and Q for 49 low-dividend firms from Fazzari, Hubbard, and Petersen [1988], (hereinafter FHP [1988]), from 1970 to 1984 compared with estimates in FHP. KZ refers to our estimates. Investment is capital expenditures (COMPUSTAT item 128). Cash flow equals the sum of earnings before extraordinary items (COMPUSTAT item 18) and depreciation (COMPUSTAT item 14). Investment and cash flow are deflated by beginning of year capital (K_{t-1}) which we define as net property, plant, and equipment (COMPUSTAT item 8). Q equals the market value of assets divided by the book value of assets (COMPUSTAT item 6). Market value of assets equals the book value of assets plus the market value of common stock less the sum of the book value of common stock (COMPUSTAT item 6) and balance sheet deferred taxes (COMPUSTAT item 74). All regressions include firm fixed effects and year effects. Standard errors are in brackets.

	KZ 1970–84	KZ 1970–84	FHP 1970–84	KZ 1970–79	KZ 1970–79	FHP 1970–79	KZ 1970–75	KZ 1970–75	FHP 1970–75
CF_t/K_{t-1}	0.395	0.500	0.461	0.477	0.578	0.540	0.558	0.634	0.670
	[0.026]	[0.023]	[0.027]	[0.035]	[0.030]	[0.036]	[0.040]	[0.034]	[0.044]
Q_{t-1}	0.039		0.0008	0.030		0.0002	0.021		−0.0010
	[0.005]		[0.0004]	[0.006]		[0.0004]	[0.006]		[0.0004]
Adj. R^2	0.584	0.548	0.46	0.649	0.627	0.47	0.764	0.753	0.55
N obs.	719	719	N.A.	476	476	N.A.	280	280	N.A.

TABLE II

SUMMARY OF ANNUAL FINANCING CONSTRAINT STATUS

Distribution of financing constraints by year for 49 low-dividend firms are from FHP [1988], from 1970 to 1984. Firm financing constraint status for each year are not financially constrained (NFC), likely not financially constrained (LNFC), possibly financially constrained (PFC), likely financially constrained (LFC), and financially constrained (FC).

	NFC Not financially constrained	LNFC Likely not financially constrained	PFC Possibly financially constrained	LFC Likely financially constrained	FC Definitely financially constrained	NFC + LNFC Not or likely not financially constrained	PFC + LFC + FC Possibly, likely, or definitely financially constrained
1970	34.0%	44.7%	14.9%	2.1%	4.3%	78.7%	21.3%
1971	38.3	34.0	17.0	10.7	0.0	72.3	27.7
1972	43.8	35.4	12.5	8.3	0.0	79.2	20.8
1973	39.6	45.8	6.3	4.2	4.2	85.4	14.6
1974	36.7	28.6	12.2	16.3	6.1	65.3	34.7
1975	30.6	42.9	14.3	8.2	4.1	73.5	26.5
1976	51.0	38.8	2.0	4.1	4.1	89.8	10.2
1977	59.2	28.6	4.1	0.0	8.2	87.8	12.2
1978	67.3	26.5	2.0	2.0	2.0	93.8	6.2
1979	61.2	26.5	10.2	2.0	0.0	87.8	12.2
1980	73.5	20.4	4.1	2.0	0.0	93.9	6.1
1981	71.4	20.4	6.1	0.0	2.0	91.8	8.2
1982	69.4	24.5	2.0	2.0	2.0	93.9	6.1
1983	69.4	24.5	2.0	4.1	0.0	93.9	6.1
1984	69.4	22.4	0.0	6.1	2.0	91.8	8.2
Total	54.5	30.9	7.3	4.8	2.6	85.3	14.7

5.2 Table II and Table III: who is actually constrained?

Read.

- Table II reports how often the archival classification places firm-years in each status bucket.
- Table III shows whether those buckets line up with standard operating and financial-health measures.

Numbers.

- NFC = **54.5%** of firm-years.
- LNFC = **30.9%**.
- Total with no sign of financing limits = **85.3%**.
- Median interest coverage drops from about **7.97** (NFC) to **1.09** (FC).
- Median slack drops from about **0.725** of capital to about **0.229**.

Interpretation. The famous low-dividend firms are usually *not* short of finance. Table III then shows that the categories are quantitatively sensible.

Why it matters. These tables are the core empirical foundation of the paper. Once they are accepted, the later regression results become hard to dismiss.

TABLE III

SUMMARY STATISTICS FOR FIRM CHARACTERISTICS BY YEARLY FINANCING CONSTRAINT STATUS

Distribution of financial variables by annual financing constraint status for 49 low-dividend firms from FHP [1988] from 1970 to 1984. Firm financing constraint status for each year is not financially constrained (NFC), likely not financially constrained (LNFC), possibly financially constrained (PFC), likely financially constrained (LFC), and financially constrained (FC). Each entry reports the median, mean, tenth percentile, ninetieth percentile, and number of observations. Investment (I_t), cash flow Q, and capital (K_{t-1}) are defined in Table I. Acquisitions (Acqs.) equals the value of purchase and pooling acquisitions. Interest coverage is the ratio of earnings before interest, taxes, and depreciation (EBITDA) to interest expense. Debt is the sum of the book value short-term and long-term debt. Total capital is the sum of debt, the book value of preferred stock, and the book value of common equity. Free divs. is the amount of retained earnings that are not restricted from being paid out as dividends. Cash is cash and marketable securities. Unused line$_t$ is the amount of unused line of credit at the end of year t. Slack is the sum of cash and unused line. Change in debt is the change in sum of the book value of short-term and long-term debt. Equity issue is the sum of the equity issued to the public and to acquisition targets.

	NFC Not fin. constr.	LNFC Likely not fin. constr.	PFC Possibly fin. constr.	LFC Likely fin. constr.	FC Fin. constr.	All firm-years
A. Investment, cash flow, growth						
I_t/K_{t-1}	0.368	0.324	0.359	0.273	0.243	0.348
	0.461	0.413	0.450	0.350	0.313	0.436
	0.159	0.159	0.122	0.073	0.068	0.127
	0.831	0.831	0.824	0.909	0.544	0.810
	393	221	52	34	19	719
Cash Flow$_t/K_{t-1}$	0.506	0.350	0.313	0.243	0.020	0.421
	0.614	0.435	0.366	0.191	−0.047	0.505
	0.209	0.104	−0.125	−0.126	−0.436	0.122
	1.075	0.871	1.084	0.528	0.366	1.007
	393	221	52	34	19	719
(Cash Flow$_t - I_t$)$/K_{t-1}$	0.110	0.026	−0.026	−0.071	−0.198	0.051
	0.155	0.022	−0.085	−0.159	−0.360	0.069
	−0.180	−0.316	−0.474	−0.642	−0.785	−0.285
	0.503	0.323	0.420	0.141	−0.076	0.442
	393	221	52	34	19	719
Q_t	1.313	1.171	1.159	1.096	1.082	1.231
	1.647	1.542	1.312	1.527	1.402	1.580
	0.809	0.755	0.793	0.734	0.795	0.785
	2.781	2.799	1.934	2.659	1.789	2.749
	393	221	52	34	19	719
Fraction firms with acquisitions in year$_t$	0.244	0.244	0.154	0.176	0.000	0.228
	393	221	52	34	19	719

TABLE III

(CONTINUED)

	NFC Not fin. constr.	LNFC Likely not fin. constr.	PFC Possibly fin. constr.	LFC Likely fin. constr.	FC Fin. constr.	All firm-years
Acqs.$_t/K_{t-1}$	0.000	0.000	0.000	0.000	0.000	0.000
	0.122	0.159	0.063	0.023	0.000	0.121
	0.000	0.000	0.000	0.000	0.000	0.000
	0.287	0.300	0.044	0.029	0.000	0.252
	388	217	52	34	19	710
Sales growth$_t$	0.211	0.150	0.123	0.136	0.008	0.180
	0.226	0.165	0.097	0.113	0.049	0.188
	0.021	−0.071	−0.136	−0.145	−0.275	−0.051
	0.484	0.385	0.319	0.338	0.305	0.452
	393	221	52	34	19	719
Inventory growth$_t$	0.199	0.117	0.144	0.063	−0.064	0.154
	0.215	0.160	0.135	0.049	−0.013	0.179
	−0.073	−0.175	−0.056	−0.499	−0.487	−0.135
	0.545	0.475	0.376	0.562	0.543	0.512
	393	221	52	34	19	719
B. Financial policy						
Interest coverage$_t$	7.971	5.886	4.203	2.836	1.093	6.406
	18.026	11.777	4.745	3.455	1.650	14.023
	2.746	1.608	0.000	0.666	0.000	1.707
	46.722	23.605	9.598	6.960	3.827	33.325
	393	221	52	34	19	719
Debt$_t$ to total capital$_t$	0.296	0.351	0.431	0.541	0.565	0.349
	0.293	0.352	0.454	0.573	0.621	0.344
	0.051	0.117	0.258	0.316	0.361	0.075
	0.526	0.585	0.689	0.791	0.912	0.585
	393	221	52	34	19	719
Dividends$_t/K_t$	0.000	0.000	0.000	0.000	0.000	0.000
	0.015	0.006	0.006	0.002	0.001	0.011
	0.000	0.000	0.000	0.000	0.000	0.000
	0.046	0.023	0.028	0.028	0.007	0.037
	393	221	52	34	19	719
Fraction of years dividends restricted	0.061	0.276	0.462	0.686	0.789	0.206
	393	221	52	34	19	719
Free divs.$_t/K_{t-1}$	0.208	0.013	0.000	0.000	0.000	0.101
	0.334	0.139	0.043	0.019	0.000	0.229
	0.004	0.000	0.000	0.000	0.000	0.000
	0.740	0.430	0.078	0.089	0.000	0.634
	247	129	34	29	15	454

TABLE III

(CONTINUED)

	NFC Not fin. constr.	LNFC Likely not fin. constr.	PFC Possibly fin. constr.	LFC Likely fin. constr.	FC Fin. constr.	All firm-years
Cash$_t/K_{t-1}$	0.331	0.150	0.150	0.077	0.085	0.168
	0.726	0.253	0.263	0.156	0.139	0.364
	0.050	0.034	0.041	0.029	0.016	0.033
	1.276	0.596	0.721	0.389	0.292	0.784
	393	221	52	34	19	719
Unused line$_t > 0$	0.723	0.652	0.654	0.529	0.579	0.683
	393	221	52	34	19	719
Unused line$_t/K_{t-1}$	0.270	0.178	0.136	0.043	0.072	0.203
	0.523	0.313	0.291	0.151	0.159	0.415
	0.000	0.000	0.000	0.000	0.000	0.000
	1.097	0.733	0.900	0.449	0.900	0.979
	393	221	52	34	19	719
Slack$_t/K_{t-1}$	0.725	0.420	0.344	0.211	0.229	0.557
	1.249	0.566	0.449	0.374	0.320	0.919
	0.217	0.118	0.059	0.044	0.001	0.126
	2.039	1.129	0.923	0.721	1.065	1.679
	393	221	52	34	19	719
Ch. debt$_t/K_{t-1}$	0.048	0.048	0.153	0.272	0.017	0.062
	0.168	0.157	0.405	0.473	0.012	0.191
	−0.304	−0.354	−0.470	−0.414	−0.546	−0.354
	0.718	0.760	0.983	1.581	0.974	0.797
	393	221	52	34	19	719
Equity issue$_t/K_{t-1}$	0.000	0.000	0.000	0.000	0.000	0.000
	0.224	0.149	0.042	0.020	0.046	0.177
	0.000	0.00	0.00	0.000	0.000	0.000
	0.634	0.419	0.044	0.000	0.256	0.455
	373	193	38	31	16	651

5.3 Table IV: ordered-logit validation of the classifications

Read. The dependent variable is the ordered financing-status category from NFC to FC.

- Higher cash flow reduces the probability of being classified as constrained.
- More debt raises it.
- Dividend restrictions raise it.
- More cash, more unrestricted retained earnings, and an unused line of credit reduce it.
- Q enters positively conditional on cash flow, which the authors interpret as "high opportunities but low internal funds" looking more constrained.

Why it matters. This is the paper's formal check that the archival coding matches ordinary balance-sheet intuition.

5.4 Table V: main full-sample result

Read. The table runs the standard fixed-effects investment equation separately by overall financial status over the full 1970–1984 period.

Key coefficients to remember.

- Full sample: about **0.395**.
- Never constrained firms: about **0.702**.
- Possibly constrained firms: about **0.180**.
- Likely constrained firms: about **0.340**.

Interpretation. The ranking goes the "wrong" way for the conventional view. The least constrained firms have the highest sensitivity.

TABLE IV

ORDERED LOGITS FOR PREDICTABILITY OF FINANCING CONSTRAINT STATUS

Ordered logits for the determination of annual financing constraint status for 49 low-dividend firms are from FHP [1988] from 1970 to 1984. Financing constraint for each year is ordered from not financially constrained (NFC), likely not financially constrained (LNFC), possibly financially constrained (PFC), likely financially constrained (LFC), to financially constrained (FC). Variable definitions are in Tables I and III. Standard errors are in brackets.

Dependent variable is financing constraint status				
Cash flow$_t$/K_{t-1}	−0.886	−1.164	−0.688	−0.839
	[0.230]	[0.256]	[0.222]	[0.235]
Q_t	0.276	0.370		
	[0.080]	[0.087]		
Debt/total capital$_t$	2.071	2.251	1.825	1.938
	[0.470]	[0.480]	[0.464]	[0.471]
Dividends$_t$/K_{t-1}	−23.039	−21.787	−22.551	−20.409
	[5.949]	[6.134]	[5.905]	[6.043]
Dividends restricted ($Y = 1, N = 0$)	1.496	1.365	1.472	1.294
	[0.213]	[0.224]	[0.213]	[0.222]
Unrestricted ret. earnings/K_{t-1}	−1.897	−1.936	−1.896	−1.956
	[0.497]	[0.513]	[0.499]	[0.513]
Cash$_t$/K_{t-1}	−1.704	−1.590	−1.675	−1.567
	[0.311]	[0.323]	[0.311]	[0.320]
Unused line of credit > 0	−0.711	−0.547	−0.758	−0.511
	[0.176]	[0.207]	[0.175]	[0.206]
_cut1	−0.252	0.608	−0.693	0.119
	[0.312]	[0.480]	[0.285]	[0.462]
_cut2	1.973	2.928	1.510	2.413
	[0.328]	[0.499]	[0.298]	[0.478]
_cut3	2.987	3.988	2.501	3.433
	[0.353]	[0.518]	[0.320]	[0.494]
_cut4	4.307	5.353	3.790	4.736
	[0.413]	[0.562]	[0.378]	[0.532]
Year dummies	No	Yes	No	Yes
Log likelihood	−645.0	−627.0	−650.6	−635.7
Pseudo-R^2	0.201	0.223	0.194	0.213

TABLE V

REGRESSION OF INVESTMENT ON CASH FLOW AND Q BY FINANCIALLY CONSTRAINED STATUS OVER ENTIRE SAMPLE PERIOD

Regression of investment on cash flow and Q for 49 low-dividend firms from FHP [1988] from 1970 to 1984. Variables are defined in Table I. Regressions are estimated for total sample and by financially constrained status where 19 firms are never financially constrained over the entire period (NFC or LNFC in every year), 8 firms are possibly financially constrained at some time (PFC in some year), and 22 firms are likely financially constrained at some time in the period (LFC or FC). Overall status is based on firm financing constraint status for each year of not financially constrained (NFC), likely not financially constrained (LNFC), possibly financially constrained (PFC), likely financially constrained (LFC), and financially constrained (FC). All regressions include firm fixed effects and year effects. Standard errors are in brackets.

	All firms N = 49	Firms never constrained N = 19	Firms possibly constrained N = 8	Firms likely constrained N = 22	Firms never/possibly constrained N = 27	Firms possibly/likely constrained N = 30
CF_t/K_{t-1}	0.395	0.702	0.180	0.340	0.439	0.250
	[0.026]	[0.041]	[0.060]	[0.042]	[0.035]	[0.032]
Q_{t-1}	0.039	0.009	0.016	0.070	0.033	0.059
	[0.005]	[0.006]	[0.049]	[0.018]	[0.006]	[0.017]
Adj. R^2	0.584	0.793	0.240	0.410	0.655	0.358
N obs.	719	279	113	327	392	440

Why it matters. If you remember one table, remember this one.

5.5 Tables VI, VII, and VIII: subperiods and objective definitions

TABLE VI

REGRESSION OF INVESTMENT ON CASH FLOW AND Q BY FINANCIALLY CONSTRAINED STATUS IN TWO SUBPERIODS TREATING FIRM-SUBPERIODS AS DIFFERENT FIRMS

Regression of investment on cash flow and Q for 49 low-dividend firms from FHP [1988] from 1970 to 1984. Variables are defined in Table I. Sample is divided into two subperiods, 1970–1977 and 1978–1984. Firm financial constraint status is determined within each subperiod. Fifty-seven firm-subperiods are never financially constrained (NFC or LNFC every year), 14 firm-subperiods are possibly financially constrained (PFC in some year), 27 firm-subperiods are likely financially constrained (LFC or FC in some year), and 15 firm-subperiods are NFC every year. Overall subperiod status is based on firm financing constraint status for each year of not financially constrained (NFC), likely not financially constrained (LNFC), possibly financially constrained (PFC), likely financially constrained (LFC), and financially constrained (FC). Regressions include firm fixed effects for each subperiod, resulting in up to 98 firm-period fixed effects, and year effects. Standard errors are in brackets.

	All firms N = 98	Firms never constrained N = 57	Firms possibly constrained N = 14	Firms likely constrained N = 27	Firms never/possibly constrained N = 71	Firms possibly/likely constrained N = 41	Firms NFC every year N = 15
CF_t/K_{t-1}	0.436	0.680	0.259	0.274	0.523	0.262	0.779
	[0.028]	[0.041]	[0.067]	[0.050]	[0.034]	[0.037]	[0.084]
Q_{t-1}	0.033	0.010	0.081	0.048	0.025	0.054	0.002
	[0.005]	[0.006]	[0.059]	[0.023]	[0.006]	[0.021]	[0.009]
Adj. R^2	0.604	0.721	0.402	0.391	0.661	0.402	0.837
N obs.	719	416	98	205	514	303	110

TABLE VII

REGRESSION OF INVESTMENT ON CASH FLOW AND Q BY FINANCIALLY CONSTRAINED STATUS OVER 1970–1977 AND 1978–1984

Regression of investment on cash flow and Q for 49 low-dividend firms from FHP [1988] from 1970 to 1984. Variables are defined in Table I. Sample is divided into two subperiods, 1970–1977 and 1978–1984. Firm financing constraint status is determined within each subperiod. Fifty-seven firm-subperiods are never financially constrained (NFC or LNFC every year), 14 firm-subperiods are possibly financially constrained (PFC in some year), 27 firm-subperiods are likely financially constrained (LFC or FC in some year), and 15 firm-subperiods are NFC every year. Overall subperiod status is based on firm financing constraint status for each year of not financially constrained (NFC), likely not financially constrained (LNFC), possibly financially constrained (PFC), likely financially constrained (LFC), and financially constrained (FC). Regressions include firm fixed effects for each subperiod. Standard errors are in brackets.

	1970–1977						
	All firms N = 49	Firms never constrained N = 23	Firms possibly constrained N = 7	Firms likely constrained N = 19	Firms never/possibly constrained N = 80	Firms possibly/likely constrained N = 19	Firms NFC every year N = 5
CF_t/K_{t-1}	0.505	0.746	0.247	0.364	0.553	0.306	0.783
	[0.037]	[0.051]	[0.086]	[0.069]	[0.045]	[0.049]	[0.142]
Q_{t-1}	0.035	0.006	0.027	0.025	0.023	0.029	0.002
	[0.017]	[0.007]	[0.082]	[0.023]	[0.007]	[0.022]	[0.015]
Adj. R^2	0.696	0.827	0.381	0.454	0.755	0.446	0.832
N obs.	378	179	50	149	229	199	40
	1978–1984						
	All firms N = 49	Firms never constrained N = 34	Firms possibly constrained N = 7	Firms likely constrained N = 8	Firms never/possibly constrained N = 41	Firms possibly/likely constrained N = 15	Firms NFC every year N = 10
CF_t/K_{t-1}	0.326	0.571	0.272	0.141	0.470	0.160	0.800
	[0.044]	[0.069]	[0.152]	[0.061]	[0.058]	[0.053]	[0.126]
Q_t	0.054	−0.019	0.154	0.413	0.007	0.272	−0.054
	[0.026]	[0.028]	[0.088]	[0.084]	[0.027]	[0.073]	[0.047]
R^2	0.392	0.467	0.422	0.459	0.439	0.402	0.703
N obs.	341	237	48	56	285	104	70

TABLE VIII

REGRESSION OF INVESTMENT ON CASH FLOW AND Q BY OTHER MEASURES OF FINANCIALLY CONSTRAINED STATUS OVER ENTIRE SAMPLE PERIOD AND ENTIRE SUBPERIOD

Regression of investment on cash flow and Q for 49 low-dividend firms from FHP [1988] from 1970 to 1984. Variables are defined in Tables I and III. Regressions (1)–(3) are estimated for total sample and by (i) whether firms ever had interest coverage below 2.5; and (ii) firms are not explicitly restricted from paying dividends over the entire sample period. Interest coverage is the ratio of EBITDA to interest expense. Regressions (4)–(6) are estimated using firm financial status over sample subperiods 1970–1977 and 1978–1984. Regressions (1)–(3) include firm fixed effects and year effects. Regressions (4)–(6) include firm fixed effects for each subperiod, resulting in up to 98 firm-period fixed effects, and year effects. Standard errors are in brackets.

	(1) All firms N = 49	(2) Firms that never have coverage below 2.5 from 1970–1984 N = 13	(3) Firms that never have restricted dividends N = 17	(4) All firms subperiods 1970–1984 and 1978–1984 N = 98	(5) Firms that never have coverage below 4.5 in subperiod N = 25	(6) Firms that never have restricted dividends in subperiod N = 56
CF_t/K_{t-1}	0.395	0.673	0.435	0.436	0.801	0.499
	[0.026]	[0.056]	[0.042]	[0.028]	[0.062]	[0.038]
Q_{t-1}	0.039	0.011	0.035	0.033	−0.003	0.027
	[0.005]	[0.008]	[0.007]	[0.005]	[0.008]	[0.006]
Adj. R^2	0.584	0.756	0.674	0.604	0.772	0.715
N obs.	719	191	247	719	189	402

Read.

- Table VI repeats the analysis at the firm-subperiod level.
- Table VII separates the 1970–1977 and 1978–1984 periods.
- Table VIII uses objective quantitative rules instead of the qualitative classification.

Main things to remember.

- In Table VI, unconstrained subperiods have a coefficient around **0.680**.
- The **15** subperiods that are NFC in *every single year* have an even higher coefficient of about **0.779**.
- In Table VIII, firms whose interest coverage never falls below **2.5** over the full sample have a coefficient around **0.673**.

- At the subperiod level, firms whose coverage never falls below **4.5** have a coefficient around **0.801**.

Interpretation. The result is not an artifact of one particular qualitative coding rule.

Why it matters. These tables are what make the paper hard to rebut by saying "your archival classification is subjective."

5.6 Table IX and Table X: lagged status and cash stock

TABLE IX

REGRESSION OF INVESTMENT ON CASH FLOW AND Q BY ANNUAL FINANCING CONSTRAINT STATUS, RESTRICTED DIVIDEND STATUS, AND LOW SLACK STATUS

Regression of investment on cash flow, Q, and cash flow interacted with financially constrained status, restricted dividend status, and low cash and unused line of credit status for 49 low-dividend firms are from FHP [1988] from 1970 to 1984. Variables are defined in Tables I and III. Firm financing constraint for each year is not financially constrained (NFC), likely not financially constrained (LNFC), possibly financially constrained (PFC), likely financially constrained (LFC), or financially constrained (FC). The noninteracted cash flow variable represents years in which firms are NFC. Regressions (1) and (2) use financial constraint status at the beginning of the fiscal year (based on status at the end of the previous fiscal year). Regression (3) interacts cash flow with a dummy variable that equals one if a firm's covenants restrict it from paying dividends in the previous fiscal year. Regression (4) interacts cash flow with a dummy variable that equals one if a firm's slack in the previous fiscal year is in the lowest quartile of firm-years (less than 0.28 of net property, plant, and equipment). Slack is the sum of cash and unused line of credit. Regressions include firm fixed effects and year effects. Standard errors are in brackets.

(1) Interact annual financial constraint status		(2) Investment by annual financial constraint status		(3) Interact annual restricted dividend status		(4) Interact annual low slack status	
CF_t/K_{t-1}	0.407	Constant	0.202	CF_t/K_{t-1}	0.358	CF_t/K_{t-1}	0.359
	[0.043]		[0.027]		[0.029]		[0.027]
CF_t/K_{t-1}	0.013	LNFC	−0.060	CF_t/K_{t-1}	−0.106	CF_t/K_{t-1}	−0.061
× LNFC	[0.035]		[0.026]	× restricted dividends	[0.052]	× low slack	[0.040]
CF_t/K_{t-1}	−0.235	PFC	−0.112				
× PFC	[0.055]		[0.045]				
CF_t/K_{t-1}	−0.382	LFC	−0.167				
× LFC	[0.086]		[0.054]				
CF_t/K_{t-1}	−0.394	FC	−0.251				
× FC	[0.162]		[0.069]				
Q_{t-1}	0.041	Q_{t-1}	0.101	Q_{t-1}	0.048	Q_{t-1}	0.051
	[0.011]		[0.011]		[0.011]		[0.011]
Adj. R^2	0.504		0.342		0.476		0.475
N obs.	674		674		674		674

TABLE X

REGRESSION OF INVESTMENT ON CASH FLOW, CASH STOCK, AND Q BY ANNUAL FINANCING CONSTRAINT STATUS

Regression of investment on cash flow, cash stock, Q, and cash flow and cash stock interacted with financially constrained status for 49 low-dividend firms from FHP [1988] from 1970 to 1984. Variables are defined in Tables I and III. Firm financing constraint status for each year is not financially constrained (NFC), likely not financially constrained (LNFC), possibly financially constrained (PFC), likely financially constrained (LFC), or financially constrained (FC). The noninteracted cash flow variable represents years in which firms are NFC. Regressions include firm fixed effects and year effects. Standard errors are in brackets.

(1) Cash stock only		(2) Cash stock and cash flow		(3) Sum of cash stock and cash flow	
$\text{Cash}_{t-1}/K_{t-1}$	0.164	$\text{Cash}_{t-1}/K_{t-1}$	0.101	$[\text{Cash}_{t-1} + CF_t]/K_{t-1}$	0.163
	[0.015]		[0.015]		[0.011]
$\text{Cash}_{t-1}/K_{t-1}$ × LNFC	0.056	$\text{Cash}_{t-1}/K_{t-1}$ × LNFC	0.014	$[\text{Cash}_{t-1} + CF_t]/K_{t-1}$ × LNFC	0.079
	[0.057]		[0.060]		[0.024]
$\text{Cash}_{t-1}/K_{t-1}$ × PFC	−0.154	$\text{Cash}_{t-1}/K_{t-1}$ × PFC	0.269	$[\text{Cash}_{t-1} + CF_t]/K_{t-1}$ × PFC	−0.037
	[0.125]		[0.129]		[0.041]
$\text{Cash}_{t-1}/K_{t-1}$ × LFC	−0.463	$\text{Cash}_{t-1}/K_{t-1}$ × LFC	0.249	$[\text{Cash}_{t-1} + CF_t]/K_{t-1}$ × LFC	−0.174
	[0.219]		[0.257]		[0.064]
$\text{Cash}_{t-1}/K_{t-1}$ × FC	−0.523	$\text{Cash}_{t-1}/K_{t-1}$ × FC	0.321	$[\text{Cash}_{t-1} + CF_t]/K_{t-1}$ × FC	−0.196
	[0.340]		[0.355]		[0.121]
		CF_t/K_{t-1}	0.342		
			[0.033]		
		CF_t/K_{t-1} × LNFC	0.076		
			[0.041]		
		CF_t/K_{t-1} × PFC	−0.222		
			[0.062]		
		CF_t/K_{t-1} × LFC	−0.384		
			[0.108]		
		CF_t/K_{t-1} × FC	−0.405		
			[0.179]		
Q_{t-1}	0.085	Q_{t-1}	0.040	Q_{t-1}	0.040
	[0.011]		[0.010]		[0.010]
Adj. R^2	0.306		0.441		0.393
N obs.	674		674		674

Read.

- Table IX uses only lagged annual financing status.
- Table X examines cash stock, cash flow, and the sum of the two.

From Table IX.

- Using lagged status, more constrained categories still have *lower*, not higher, cash-flow sensitivities.
- In an investment-level regression, investment declines monotonically as lagged financing status worsens.
- The paper highlights one concrete number: after LFC years, investment is roughly **0.17** lower than after NFC years, conditional on Q.

From Table X.

- When internal funds are measured as $\text{Cash}_{t-1} + \text{CF}_t$, the base sensitivity for NFC years is about **0.163**.
- The interaction terms become more negative as status worsens:
  - LNFC: +0.079,
  - PFC: −0.037,
  - LFC: −0.174,
  - FC: −0.196.

Interpretation. Even when one broadens the liquidity measure beyond current cash flow, the least constrained firms remain the most responsive to internal funds.

5.7 Table XI and Table XII: outliers and the distress objection

TABLE XI

REGRESSION OF INVESTMENT ON CASH FLOW AND Q BY FINANCIALLY CONSTRAINED STATUS OVER ENTIRE PERIOD WITHOUT HIGH SALES GROWTH OR HIGH INVESTMENT GROWTH OBSERVATIONS

Regression of investment on cash flow and Q for 49 low-dividend firms from FHP [1988] from 1970 to 1984. Variables are defined in Table I. Regressions are estimated for total sample and by financially constrained status, where 19 firms are never financially constrained over the entire period (NFC or LNFC in every year), 8 firms are possibly financially constrained at some time (PFC in some year), and 22 firms are likely financially constrained at some time in the period (LFC or FC). Overall status is based on firm financing constraint status for each year of not financially constrained (NFC), likely not financially constrained (LNFC), possibly financially constrained (PFC), likely financially constrained (LFC), and financially constrained (FC). All regressions include firm fixed effects and year effects. Standard errors are in brackets.

	No firm-years with more than 30% sales growth				No firm-years with investment exceeding initial capital (K_{t-1})			
	All firms N = 49	Firms never constrained N = 19	Firms possibly constrained N = 8	Firms likely constrained N = 22	All firms N = 49	Firms never constrained N = 19	Firms possibly constrained N = 8	Firms likely constrained N = 22
CF_t/K_{t-1}	0.246	0.531	0.104	0.233	0.203	0.366	0.149	0.211
	[0.050]	[0.124]	[0.045]	[0.058]	[0.031]	[0.042]	[0.046]	[0.032]
Q_{t-1}	0.051	0.033	0.048	0.049	0.046	0.023	−0.001	0.067
	[0.012]	[0.014]	[0.053]	[0.024]	[0.009]	[0.010]	[0.027]	[0.013]
Adj. R^2	0.328	0.502	0.155	0.270	0.449	0.597	0.252	0.427
N obs.	535	201	79	255	679	263	109	307

TABLE XII

MEDIAN FIRM CHARACTERISTICS BY FINANCIALLY CONSTRAINED STATUS IN ENTIRE SAMPLE PERIOD

Median firm characteristics by overall financial status for 49 low-dividend firms from FHP [1988] from 1970 to 1984. Overall status is based on firm financing constraint status for each year of not financially constrained (NFC), likely not financially constrained (LNFC), possibly financially constrained (PFC), likely financially constrained (LFC), and financially constrained (FC). For the entire period, 19 firms are never financially constrained over the entire period (NFC or LNFC in every year), 8 firms are possibly financially constrained at some time (PFC in some year), and 22 firms are likely financially constrained at some time in the period (LFC or FC). Each entry reports the median and number of observations. Investment (I_t), cash flow, Q, and capital (K_{t-1}) are defined in Table I. Interest coverage is the ratio of earnings before interest, taxes, and depreciation (EBITDA) to interest expense. Debt is the sum of the book value of short-term and long-term debt. Total capital is the sum of debt, the book value of preferred stock, and the book value of common equity. Free divs. is the amount of retained earnings that are not restricted from being paid out as dividends. Cash is cash and marketable securities. Unused line$_t$ is the amount of unused line of credit at the end of year t. Slack is the sum of cash and unused line.

	Never constrained N = 279	Possibly constrained N = 113	Likely constrained N = 327	All firm-years N = 719
A. Investment, cash flow, growth				
I_t/K_{t-1}	0.348	0.403	0.337	0.348
Cash Flow$_t$/K_{t-1}	0.451	0.517	0.364	0.421
(Cash Flow$_t$ − I_t)/K_{t-1}	0.081	0.142	0.001	0.051
Q_t	1.262	1.438	1.200	1.231
Sales growth$_t$	0.194	0.176	0.172	0.180
B. Financial policy				
Interest coverage$_t$	8.070	9.928	4.842	6.406
Debt$_t$ to total capital$_t$	0.289	0.249	0.415	0.349
Fraction of years dividends restricted	0.115	0.070	0.327	0.206
Free divs.$_t$/K_{t-1}	0.186	0.315	0.023	0.101
Cash$_t$/K_{t-1}	0.215	0.239	0.109	0.168
Unused line$_t$ > 0	0.631	0.649	0.730	0.683
Unused line$_t$/K_{t-1}	0.153	0.208	0.256	0.203
Slack$_t$/K_{t-1}	0.626	0.630	0.481	0.557
Ch. debt$_t$/K_{t-1}	0.048	0.000	0.094	0.062
Years with equity issue	0.234	0.167	0.189	0.203

Read.

- Table XI trims very high-growth observations.
- Table XII compares median firm characteristics by overall financing status.

Main points from Table XI.

- Excluding firm-years with sales growth above 30% or with huge jumps in property, plant, and equipment lowers the overall coefficients.

- But the *ordering* remains: unconstrained firms still have the highest sensitivities.
- The trimmed full-sample sensitivity of the low-dividend sample falls to roughly **0.20–0.25**, close to the FHP high-dividend benchmark of about **0.23**.

Main points from Table XII.

- Never constrained firms: median interest coverage about **8.07**.
- Possibly constrained firms: median interest coverage about **9.93**.
- Likely constrained firms: median interest coverage about **4.84**.

Interpretation. The findings are not driven by a few explosive-growth years, and the middle categories are not simply collections of distressed firms.

6 Short summary

- The paper asks whether investment–cash flow sensitivities are valid measures of financing constraints.
- In theory, they are not guaranteed to rank firms monotonically by financing frictions.
- In the classic FHP low-dividend sample, most firm-years are not actually constrained once one reads the firms' own disclosures and financial statements.
- Empirically, the least constrained firms have the *highest* sensitivities of investment to internal funds.
- This result survives subperiod splits, objective financial-health cutoffs, lagged-status regressions, Euler-equation tests, cash-stock specifications, and outlier trimming.
- The paper's core message is that high investment–cash flow sensitivity cannot by itself be interpreted as evidence of tighter financing constraints.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes two things.

- **Theoretically,** the monotonicity assumption behind the investment–cash flow literature is not generally valid.
- **Empirically,** in the most famous sample used to support that literature, firms that appear less constrained have higher investment–cash flow sensitivities than firms that appear more constrained.

So the investment–cash flow coefficient is not a reliable ordinal measure of financing constraints.

7.2 Economic intuition

There are at least two broad ways to understand the result.

- One possibility is that the shape of the external-finance cost function and the shape of the investment technology make the sensitivity nonmonotonic in a standard optimizing model.
- Another possibility is behavioral or organizational: some successful firms prefer to finance investment out of internal funds even though cheap external finance is available.

The paper does not fully resolve which mechanism is dominant, but it makes clear that the usual reduced-form interpretation is too simple.

7.3 Takeaway

This is a critique paper with a very strong message:

> A high investment–cash flow sensitivity is evidence that internal funds matter for investment, but it is *not* evidence by itself that a firm is more financially constrained than another firm.

That is why the paper matters so much. It does not deny financing frictions; it changes how one is allowed to infer them from data.